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Forecasting Price of Cryptocurrency using time series data and machine learning algorithms

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Abstract- *The cryptocurrency market is a rapidly changing environment that presents considerable problems for investors, traders, and analysts. Predicting cryptocurrency prices has been a popular topic in recent years. These currencies are used by millions of people worldwide, and their value fluctuate according to demand. Researchers and investors are making progress in constructing predicting models for cryptocurrency values, thanks to increased availability of historical data and breakthroughs in machine learning techniques. In this study we took four different models (Auto regressive integrated moving average, Support Vector Regression, Long Short Term Memory and Gated Recurrent Unit) to find the most effective way to predict price of cryptocurrency and then deploying into a web application for users to predict price. The study compares the mean absolute percentage error of these three models. For our study we have taken Litecoin a well-known cryptocurrency for model training and testing. The data has been fetched from yahoo finance which then goes through transformation and finally fitted into the ml model. The most effective model turns out to be LSTM with a MAPE of 37.45 to produce a prediction.*

Keywords- RNN- Recurrent Neural Network, LSTM-Long Short Term Memory, SVR- Support Vector Regression, ARIMA- Auto regressive integrated moving average, GRU Gated Recurrent Unit, MAPE-Mean Absolute Percentage Error, RMSE- Root mean square error

I. INTRODUCTION

Cryptocurrencies have surfaced as a new and quickly expanding asset class that has piqued the interest of both buyers and academics. However, forecasting the future fluctuations in the prices of cryptocurrencies has become difficult due to their extremely unpredictable character and absence of governmental supervision.[1] Forecasting models have received significant notice in recent years as a means of giving insight into the future patterns of coins. The purpose of this study paper is to investigate the efficacy of various projection techniques for determining the values of various coins. This research will add to a better grasp of the variables that impact cryptocurrency values by analyzing the precision and dependability of these models, and will provide useful insights for buyers and lawmakers equally. These cryptocurrencies has a potential to disrupt other online payments like cards and Paytm wallets.

There are various methods used for cryptocurrency price forecasting, including time series analysis, machine learning, and deep learning techniques The process of analyzing past price data to find patterns and trends that can then be used to predict future values is known as time series analysis. Machine learning and deep learning methods involve teaching algorithms on large databases of past price and other pertinent data to spot complicated relationships and patterns that conventional statistical methods find difficult to discern. [2-8]

Numerous studies have been conducted in recent years that have applied various predicting methods to bitcoin market data, with differing degrees of success. Some studies reported excellent model accuracy rates, while others reported much lower rates. The choice of model, the quantity of data used, the time span examined, and market circumstances are all factors that impact the precision of predicting models.[9]

In order to make an effective model study of different models is necessary on different cryptocurrency. Thus, in order to achieve this we studied four different types of models- ARIMA, LSTM, GRU and SVR and measured accuracies. The best model was then chosen and then deployed on a web application made with streamlit.

The cryptocurrency used in this project is Litecoin. Many research work had been done on popular cryptocurrencies such as bitcoin, Ethereum etc but as the market evolve and new techniques are invented no one knows exactly when a cryptocurrency is going to be in it's highest demand. We chose this as not much research work is done on this specific cryptocurrency. It's prices were taken from 2015 to 2023 to train and test our models. This research work can be used to combine the two or models to extract the goodness of different models and predict a desired output.

Existing types of models and motivation behind this project.

Although the data available for these cryptocurrency is enough but previously implemented models are trained on very less data. Previously the work was done using autoregressive integrated moving average (ARIMA) model [11], SVR model for bitcoin[12].

Till now from the best of our knowledge no one has trained and compared the efficiency of these four models on the latest dataset for Litecoin. In addition to this we also host it on a web application.
We are building this project to make it easier for new investors and traders to enter into this market. The motivation behind this is to create an interactive website which when deployed can deliver results in a much better and in a concise way. The challenges not only lies in predicting the values but also on which parameters to consider, the layers of RNN used such that the model is not slow and can deliver results for a longer duration with much confidence.

Here are some advantages of the data generated via this model:

- **Profitable trading**: Cryptocurrency markets are known for their high volatility, which presents opportunities for profitable trading. By accurately predicting the prices of cryptocurrencies using LSTM, traders can make informed investment decisions and maximize their profits.

- **Risk management**: Cryptocurrencies are often associated with high risk due to their unpredictable price fluctuations. LSTM can be used to forecast future prices and help investors and traders manage their risk exposure.

- **Investment strategy**: LSTM can be used to develop investment strategies for cryptocurrency portfolios. By predicting the future prices of various cryptocurrencies, investors can make informed decisions about which assets to hold and when to buy or sell them.

- **Understanding market behavior**: LSTM can help to identify patterns and trends in the cryptocurrency market. By analyzing historical price data, LSTM can reveal insights into market behavior and inform investment decisions.

Overall, the ability to predict cryptocurrency prices using LSTM can provide investors and traders with valuable insights and help them to make informed decisions in a rapidly evolving market.
The results are quite promising and can be used by the people.

We divided our study into three main parts: Literature Survey, proposed work, and discussion, In literature survey we go through the existing research conducted on the prediction of price of cryptocurrency. In proposed work we applied our method for predicting values using four different models- ARIMA, SVR, LSTM and GRU. The best of them is selected and deployed on web application. In the results and conclusion section we wrote our observation and performance of the model after training and testing on Litecoin dataset. In addition to this we also discuss about future works and improvement of accuracy of this model.

II. LITERARY SURVEY

Vapnik (1995) introduced the support vector machine (SVM), which has been effectively used for forecasting, clustering, and classification. Support vector regression (SVR) is suggested in this research as a method for predicting Chinese real estate prices. The goal is to assess SVR's viability for predicting real estate prices. In order to accomplish the goal, five indicators are chosen as input variables, and the real estate market serves as the SVR's output variable. The SVR model is built using the quarterly data from 1998 to 2008 as the data collection. Future real estate values are predicted and examined using the scenarios. SVR model and BPNN model predicting abilities were contrasted. The experimental findings show that the SVR model outperforms the BPNN model in terms of the mean absolute error (MAE), the mean absolute percentage error (MAPE), and the root mean squared error (RMSE), and that the SVR-based approach was a useful tool for predicting real estate prices.[13]
To compute the longer-term share prices, the prediction of stock value is a challenging task that requires a solid algorithmic foundation. It will be challenging to predict expenses because stock prices are correlated with market nature. The suggested algorithm uses market data to forecast share prices using machine learning methods like a recurrent neural network called Long Short Term Memory. Weights are adjusted for each data point using stochastic gradient descent during this process. Compared to the stock price predictor algorithms currently in use, this system will deliver accurate results. With different sizes of input data, the network is trained and assessed to drive the graphical results.[14]
In order to forecast the closing price in the short and medium terms, respectively, we suggest a trading strategy tailored for the Moroccan stock market, based on two deep learning models: LSTM and GRU. Based on the forecasts provided by the two models, decision rules for buying and selling stocks are implemented, and over four 3-year periods, we simulate transactions using these decision rules with various settings for each stock. An expected return will be calculated using the returns received.[15]
Since its launch in 2009, cryptocurrency has drawn interest from both speculators and academics. Researchers have been very interested in and have conducted numerous studies on cryptocurrencies due to their substantial growth in market capitalization, variety, and volatility. Most research, however, focuses on the extremely well-liked bitcoin. The most recent study demonstrates that institutional investors have shown interest in Litecoin, despite it being in fourth position with a market cap of 2.74% as of August 1st, 2019. This represents a milestone because it is the first time that cryptocurrency and fiat money have been treated equally. This significant development is credited to Litecoin's distinctive features of low mining costs, real-time transaction speeds, and a strong, cost-effective blockchain-based architecture that can handle higher volumes than bitcoin. Data analysis is done using two models, Auto-Regressive Integrated Moving Average (ARIMA) and Long Short-Term Memory (LSTM), to determine the most accurate model for price forecast. Data gathered over a period of five and half years, from 2014 to 2019, is analysed, and the efficiency of both models is assessed.[16].

III. PROPOSED METHODOLOGY

With our work, we aim to create a project that will enable to track and predict prices with a greater accuracy and help investors in making decisions profitable.
To solve this and handle enormous amount of data, training through various parameters is necessary. In order
to have the best result we need to consider different approaches to predict the price. In our research we will consider four different models namely ARIMA (Auto Regressive Integrated moving average), SVR (Support Vector Regression), LSTM (Long Short Term Memory) and GRU (Gated Recurrent Unit) and train them to find out which model leads to best result.

Before fitting the values, we have to make convert the time series values into a data structure such that we can feed the values into the model. We took 100 days of values and corresponding to that 101th value was predicted. The same was inserted back into the model and subsequently other values were predicted.

A. Proposed Flowchart

The flow of our project has been depicted in the following flowchart.

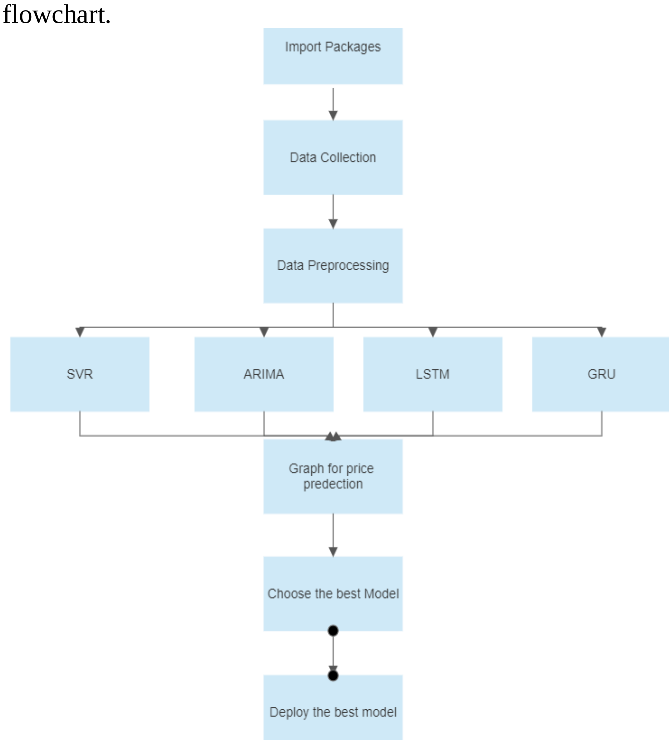

Fig. 3.1 Process Flowchart

B. Dataset Description

For our research we chose to train and test our dataset on Litecoin. This data was fetched from yahoo finance. The consists of total 2739 values starting from 1st September 2015 till 2nd March 2023. The dataset was taken with a specific goal in mind which was to provide most effective model to produce predicted values. We then transformed the data into the values lying in the range of 0 and 1. The regression techniques like SVR, ARIMA, LSTM and GRU were then applied to achieve the results.

	Open	High	Low	Close	Adj Close	Volume
count	2739.000000	2739.000000	2739.000000	2739.000000	2739.000000	2.739000e+03
mean	75.618825	78.567851	72.421642	75.632598	75.632598	1.657823e+09
std	66.733300	70.137408	62.955395	66.682585	66.682585	2.256211e+09
min	2.638140	2.724670	2.579970	2.633140	2.633140	5.074800e+05
25%	31.574616	32.922447	30.415462	31.668301	31.668301	1.883470e+08
50%	57.639591	59.180298	56.001781	57.647881	57.647881	6.367447e+08
75%	108.747696	112.303493	105.159203	108.797718	108.797718	2.590185e+09
max	387.869171	412.960144	345.298828	386.450775	386.450775	1.799426e+10

Fig 3.2 Shows description of dataset

C. Data Pre-processing

Data preprocessing is necessary for the removal of null values and fitting the data in a correct shape for the specific ml model. For preprocessing of data first the data was transformed into values lying between 0 and 1 for accurate predictions. Once transformed the data was split into test and train data. 70% of the total data was used for training and rest of the data used for testing. The model was trained so as after looking 100 days of data 101th data was predicted. This is called lookback and here we set it to 100. To make our data in such a way we took two lists of shape (X,100,1) and (X,1) and fed it into the input layer of the model. This method helped us to achieve accuracy which is better than the previous research work done considering the size of the dataset.

D. Algorithm

1. Support vector Regression

This is a type of machine learning algorithm which is widely used for regression problems, where he goal is to predict a continuous output variable. basic idea behind SVR is to find a hyperplane in the high-dimensional feature space that best separates the data into two classes: those points that lie within the margin and those that lie outside it. These subset of datapoints are called support vectors and the main aim is to minimize the amount of errors. Once the hyperplane has been defined, SVR uses them to make predictions for new data points.

In our case, the utilization of SVR for prediction of prices for Litecoin resulted in MAPE of 256 which is considered to be below average. So this cannot be considered for our primary model to be deployed.

2. Auto Regressive integrated moving average

ARIMA is a time series forecasting method that consists of three parts: autoregression, differencing, and moving average. Autoregression predicts future values by using past values, whereas differencing eliminates patterns and moving average model mistakes. ARIMA is a strong algorithm, but it is susceptible to noise and may necessitate cautious parameter adjustment.

In our case, the utilization of ARIMA for prediction of prices for Litecoin resulted in MAPE of 92 which is considered to be good in terms of size of the dataset. So, this can be considered for our primary model to be deployed.

3. Gated Recurrent Unit

GRU is a recurrent neural network that is widely used in NLP. It has two gates to regulate information flow: update and restart. It is comparable to LSTM, but it has fewer parameters and trains quicker. GRU is capable of dealing with variable-length patterns and capturing long-term relationships. It is frequently used in activities such as voice detection, sentiment analysis etc.

The utilization of GRU for prediction of prices for Litecoin resulted in MAPE of 223 which is considered to be average in terms of size of the dataset. We won't be considering this for deployment because of fewer parameters it considers while training and high mape.

4. Long Short Term Memory

LSTM is a type of recurrent neural network (RNN) architecture that is designed to avoid the vanishing gradient problem often encountered in traditional RNNs. This means in artificial neuron when an input is supplied it generates an output by processing the current input and the weights for the previous outputs generated by previous neurons. This makes this technique different as there is a recurrence of produced output between the hidden layers. This method helps to remember on what context the model is training through and thus produces results which are very close to actual values.
While this technique is useful it has some shortcomings. One of them is to handle the output when the data is enormous as the one which we are using. This is known as long term dependency problem and in order to solve this we use LSTM.
LSTM provide an internal state in the neuron. The internal state is nothing but cells which consists of three different gates: Forget gate, input Gate and Output gate.
Input Gate: The input gate determines how much new information should be added to the cell state. The information is stored in 0's and 1's that determines in our model how much new information is added from past years of data.
Output Gate: The output gate determines how much information should be output from the cell.
Forget Gate: The forget gate determines how much old information should be discarded from the cell state. It consists of a sigmoid layer, which outputs a value between 0 and 1 that determines how much of the previous cell state should be retained.

The utilization of LSTM for prediction of prices for litecoin resulted in MAPE of 37 which is considered to be excellent in terms of size of the dataset. We will be considering this for deployment and predict prices of cryptocurrency.

IV. IMPLEMENTATION

Our project has been divided into two modules:

- Building model for the prediction
- Deploying it on a webapp using streamlit

A. Building model for prediction

For building and training the data goes through multiple stages. In the first step we import dataset. All the necessary parameters are taken care of, and all others ae removed so that the model can become efficient. Further process includes more pre-processing and the use of min-max scaler comes into play. The min-max scaler is a data normalization technique used to scale and transform data values to fit within a specified range. This technique works by transforming the data values to be between 0 and 1, where 0 represents the minimum value of the data and 1 represents the maximum value of the data. The next stage involves splitting of data into test and train. Over 70% of data was used for training and rest for testing.
For the model building part we used four layers of LSTM. Reason for this is to consider all the non-linearity as LSTM are good with handling the non-linearity in the data. Initially in the first layer we took 50 neurons in our cell. Then the count was increased to 60, 80 and 120 subsequently. Dropout layers are also added in between to ensure that there is no overfitting of data. Dropout randomly drops some neurons and hence an optimal number is ensured such that there is minimal loss of valuable parameters yet efficient in predicting the values. At the end dense layer is added so that all the layers are connected and the model is fully functional. The model's EPOCS were set to 200.
The model once trained will predict the values for other cryptocurrencies.

B. Deploying model on a web app

After successfully building the model, we deploy the same by using streamlit. It is an open-source python framework which helps to quickly create web application

for data science and machine learning in short time.
Streamlit offers variety of customizations according to the one's need.

User can see their current price on the application, predict the price for next 10 days with high confidence.

V. RESULTS

After successful implementation of the models, we in our study found that for Litecoin the best model turns out to be LSTM. We used MAPE or Mean Absolute Percentage Error for comparing which is defined as-

$$\mathrm{M} = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{A_t - F_t}{A_t}\right| \tag{1}$$

Where A_t is actual value and F_t is forecast value

The formula calculates the absolute percentage difference between the actual and forecasted values for each observation, and then takes the average of these absolute percentage differences across all observations. The MAPE is expressed as a percentage.

The following table shows the MAPE with models-

Model	MAPE %
SVR	256.800
ARIMA	92.636
GRU	236.521
LSTM	**37.450**

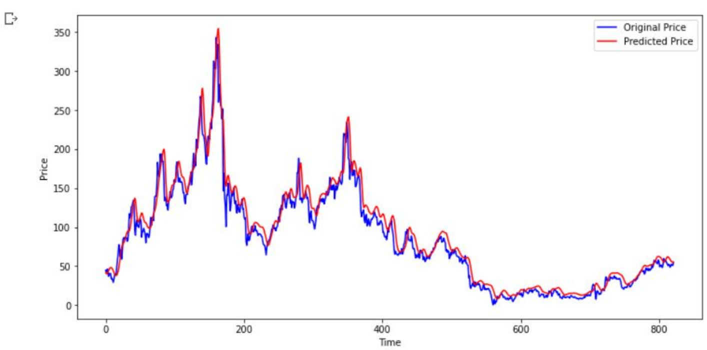

Fig 5.1 Graph showing actual price vs predicted price from LSTM model for Litecoin

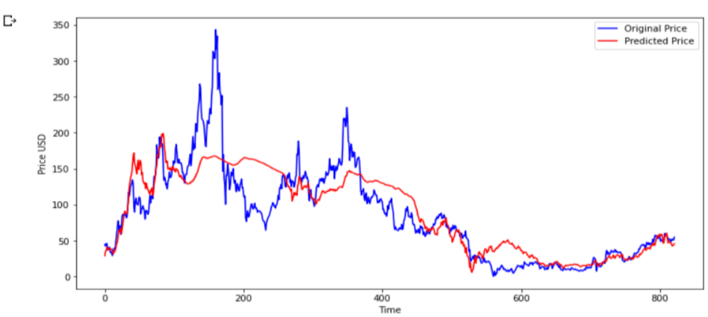

Fig 5.2 Graph showing actual price vs predicted price from SVR model for Litecoin

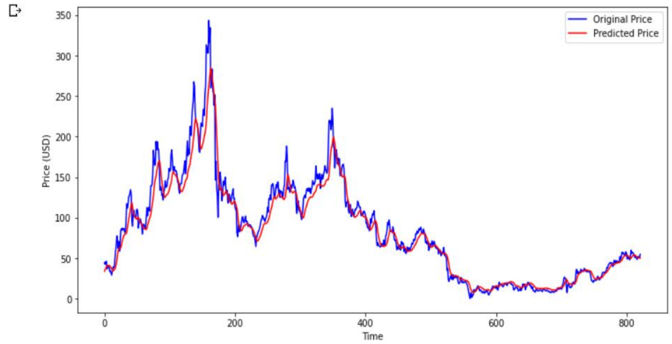

Fig 5.3 Graph showing actual price vs predicted price from GRU model for Litecoin

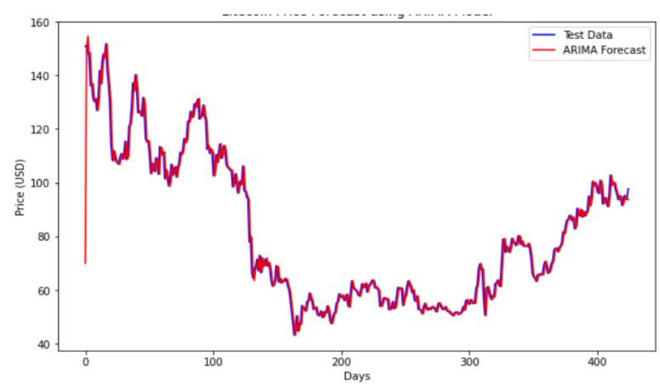

Fig 5.4 Graph showing actual price vs predicted price from ARIMA model for Litecoin

A web application has been hosted which can be accessed from anywhere.

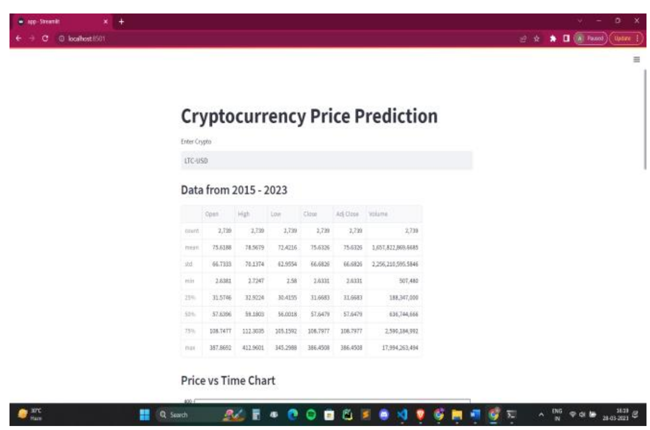

Fig 5.5 Shows deployed model on Streamlit

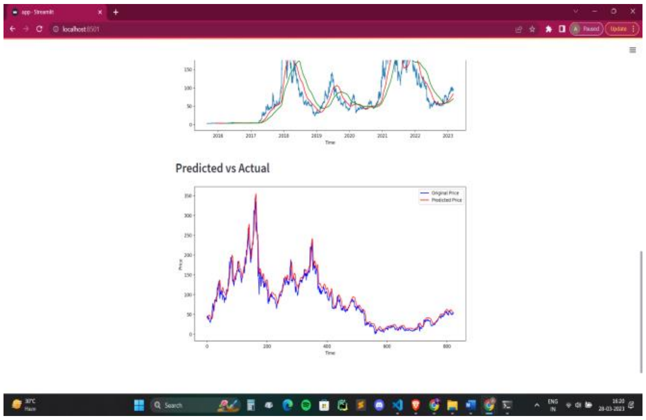

Fig 5.6 Shows Actual vs Predicted price on Streamlit

VI. CONCLUSION AND FUTURE WORKS

This study to predict prices of cryptocurrency by comparing four different models – SVR, ARIMA, GRU and LSTM were successfully implemented and the best among them was found to be LSTM with MAPE of 37.45. The least effective model is support vector regression. These results for Litecoin shows that prediction using LSTM is more likely to produce good results than other methods and similar is expected for the other cryptocurrency aswell.

Considering the best model i.e LSTM the rmse we get is 12.27 which is better than the values 150.09 in [17] and 183.84 in [18] considering the size of dataset and the currency used.

For further improvement in accuracy more complex models and integration of the best models (GRU and LSTM) can be done. Addition of weights by sentiment analysis is also a great scope as external environment has a major effect when it comes to prediction of prices. The best example of this is seen dogecoin where it can be seen a tweet from Elon Musk can change the price of the currency.

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